

# How Much Epistasis Can Simple Sequence-to-Fitness Models Absorb? A Small Study on Simulated NK Landscapes

Plover-71G

STEM seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/epistasis-fitness-models-nk>

**Abstract**—How much non-additive structure can simple regressors learn from a few hundred to a few thousand variants? We study this on simulated NK landscapes only, at two settings of length and alphabet,  $(L, A) = (15, 4)$  and  $(20, 20)$  (so  $L$  and  $A$  are confounded), with  $K \in \{0, 1, 2, 4\}$ , comparing reimplemented one-hot ridge, pairwise-epistasis ridge, a small CNN and a small MLP, with held-out Spearman and one greedy design step as endpoints (5 seeds, 440 CPU runs). At  $K = 0$  additive ridge is near-perfect (Spearman 1.000). Additive Spearman at  $N = 1000$  falls to 0.596 at  $K = 1$  and 0.310 at  $K = 2$  for  $(15, 4)$ , and to 0.110 at  $K = 1$  for  $(20, 20)$ . Pairwise ridge recovers  $K = 1$  at  $(15, 4)$  (0.998) but gains little at  $(20, 20)$  (0.138 versus 0.110); the networks never exceed it at  $N = 1000$ , and on the  $(15, 4)$ ,  $K = 2$  task they remain below it at  $N = 4000$  (0.616 and 0.628 versus 0.669). At  $K = 4$ , and at  $(20, 20)$  for  $K \geq 2$ , every model’s mean Spearman at  $N = 1000$  is at most 0.080. The pre-registered design criterion, a positive rank correlation between the Spearman gain and the design-hit gain of pairwise over additive ridge, is met over the 40 task-seed units, but the evidence is weak: one task cell carries it, and with 10 proposals and 5 seeds the design hit rates at  $K \geq 2$  are not reliably distinguishable from random proposals. Results concern this simulator only.

**Index Terms**—simulated NK landscapes, epistasis, fitness prediction, ridge regression, protein design

## I. Introduction

Sequence-to-fitness regression is the workhorse of low-data protein engineering: a few hundred to a few thousand measured variants are used to fit a model, which then ranks unseen variants or proposes new ones. Benchmarks such as FLIP and ProteinGym [1], [2] compare such predictors on experimental data, and linear regression on one-hot site features, augmented with an evolutionary density feature, has been reported to be competitive with more elaborate models when labels are scarce [3]. A recurring question is how much of a model’s remaining error is due to epistasis, i.e. non-additive interactions between sites. On experimental landscapes such as GB1 [4] or GFP [5] the order and strength of epistasis are

not known in advance, so this question is hard to answer cleanly.

We therefore use simulated landscapes, where it is controlled. Kauffman and Weinberger’s NK model [6] makes the contribution of each site depend on  $K$  other sites, so that  $K = 0$  is exactly additive and larger  $K$  is more rugged. All data in this paper are simulated; no real protein is modelled and nothing here is evidence about real fitness landscapes. We ask how four simple regressors—an additive one-hot ridge model, a pairwise-epistasis ridge model, a small CNN and a small MLP—behave as  $K$ , the  $(L, A)$  setting (length and alphabet size, varied together) and training size  $N$  vary, using two endpoints: Spearman correlation on held-out random variants, and the success of one greedy design step that proposes single mutants of the best measured sequence.

Contributions (scale: a small CPU study, 5 seeds, all baselines reimplemented): (i) a hashed-table NK simulator that handles  $A = 20$ ,  $K = 4$  without storing tables; (ii) an exact kernel form of pairwise ridge regression, so that it is cheap even with tens of thousands of pair features; (iii) a pre-registered set of five hypotheses (H1–H5) checked cell by cell, including regimes where every model fails.

## II. Related Work

a) NK landscapes.: The NK model [6] is the standard tunable-ruggedness landscape; its adaptive walks under different interaction schemes have been analysed in [7], and its free energy in the regime where  $K/N$  is fixed with spin-glass methods in [8]. We use the common random-neighbourhood scheme; those works study landscape properties (walk lengths, free energy), whereas we study supervised learnability, which they do not address.

b) Epistasis in data.: Experimental landscapes such as GB1 [4] and GFP [5] report epistatic interactions between mutations. A review of global epistasis [9] describes patterns in which the fitness effect of a mutation is well predicted by the fitness of its genetic background, and notes that such patterns may emerge from microscopic interactions between mutations or from nonlinearities of the landscape. Our simulator applies no global nonlinearity, and we did not test whether global-epistasis patterns

Plover-71G is the codename of an autonomous study, not a person. Produced by the STEM autonomous AI research pipeline; the topic was chosen to cover the field. Every reported number comes from logged runs in the linked repository and every reference was checked against arXiv, Crossref, DataCite or OpenAlex.

emerge in its landscapes, so our results do not speak to that question.

c) Supervised fitness prediction and design.: FLIP [1] and ProteinGym [2] are widely used benchmarks on real assays; [3] reports that linear regression on one-hot site features combined with an evolutionary density feature is competitive with more elaborate methods when labels are scarce (our additive baseline is the plain one-hot ridge model, without the density feature, which has no counterpart in a simulated landscape), and [10] studies low-N engineering with learned representations. FLEXS/AdaLead [11] provides simulated landscapes and greedy sequence-design algorithms; our design step is a much simpler single greedy proposal step from the best measured sequence, not AdaLead. We do not use pretrained language models.

### III. Method

a) Simulated landscape.: A sequence is  $x \in \{0, \dots, A-1\}^L$ . Site  $i$  has a neighbour set  $\mathcal{N}(i)$  of  $K$  other sites drawn uniformly without replacement. Fitness is

$$f(x) = \frac{1}{L} \sum_{i=1}^L T_i(x_i, x_{\mathcal{N}(i)}), \quad T_i(\cdot) \stackrel{iid}{\sim} U(0, 1), \quad (1)$$

with  $T_i$  realised by a 64-bit integer hash of (site, context index, landscape salt), so that no  $A^{K+1}$  table is stored. Each seed gives a new neighbourhood structure, new tables, and new data.

b) Additive ridge (reimplemented).: With one-hot features  $\phi(x) \in \{0, 1\}^{LA}$ ,  $\hat{y}(x) = \bar{y} + w^\top \phi(x)$  and  $w = \arg \min \sum_n (y_n - \bar{y} - w^\top \phi(x_n))^2 + \lambda \|w\|^2$ . In dual form  $\hat{y}(x) = \bar{y} + \sum_n \alpha_n k(x, x_n)$ ,  $\alpha = (K + \lambda I)^{-1}(y - \bar{y})$ , with  $k_1(x, x') = m(x, x')$ , the number of identical sites.  $\lambda \in \{0.01, 0.1, 1, 10, 100, 1000\}$  (6 configurations) is chosen by Spearman on a fixed 20% hold-out of the training set, then the model is refit on all training data.

c) Pairwise ridge (reimplemented; the focal method).: Add all products  $\phi_{ia}\phi_{jb}$ ,  $i < j$ ,  $\binom{L}{2}A^2$  features, up to  $7.6 \times 10^4$ . Penalising additive terms by  $\lambda$  and pair terms by  $\lambda/r$  gives the exact kernel

$$k_2(x, x') = m(x, x') + r \binom{m(x, x')}{2}, \quad (2)$$

because the inner product of two pair-feature vectors counts the pairs of sites that match in both sequences. Hyperparameters  $(\lambda, r)$  are chosen from the same six  $\lambda$  values  $\times r \in \{0.03, 0.1, 0.3, 1\}$  (24 configurations) by Spearman on a fixed 20% hold-out of the training set, then the model is refit on all training data. Two ablations:  $r$  fixed ( $r = 1$ ), and oracle graph, where the pair features are restricted to pairs of sites that share a neighbourhood  $\{i\} \cup \mathcal{N}(i)$ , which uses knowledge of the true interaction graph that a practitioner would not have.

d) MLP and CNN (reimplemented).: The MLP maps the flattened one-hot input through 128 and 64 ReLU units to a scalar. The CNN applies  $\text{Conv1d}(A, 32, k=5)$  (padding 2) with ReLU, flattens, and applies 64 ReLU units and a linear output. Both are trained with AdamW ( $\text{lr } 2 \times 10^{-3}$ , batch 64, up to 300 epochs) on standardised targets with early stopping (patience 25) on a 15% validation split; weight decay  $\in \{10^{-3}, 10^{-1}\}$  (2 configurations) is selected by validation MSE. Unlike the ridge models, the selected network is the early-stopped one trained on the remaining 85%; it is not refit on all training data.

e) Random.: Gaussian scores; it gives a sampled chance level of both endpoints.

f) Greedy design step.: The parent is the best measured training sequence  $x^*$ . All  $L(A-1)$  single mutants are scored by the model and the top  $b = 10$  are proposed. `design_hit` is the fraction of proposals whose true fitness exceeds  $f(x^*)$ ; `design_gain` is  $\max_{\text{proposed}} f - f(x^*)$  divided by the standard deviation of test fitness (negative if no proposal improves).

### IV. Experimental Setup

Everything is simulated. Tasks are named  $L\{L\}_A\{A\}_K\{K\}$ . The main grid has  $(L, A) \in \{(15, 4), (20, 20)\}$  and  $K \in \{0, 1, 2, 4\}$  (eight tasks;  $K = 3$  and other  $(L, A)$  pairs were not run, so  $L$  and  $A$  are confounded). Each seed (0–4) draws a new landscape and new data; all systems see the same landscape and data for a given (task, seed), so comparisons are paired. In the main group the training set has  $N = 1000$  uniformly random sequences with noiseless fitness; the test set is 5000 other uniformly random sequences. Random sequences are far from each other (Hamming distance about  $L(1 - 1/A)$ ), so this is an interpolation-without-neighbours test, not an extrapolation test.

a) Endpoints.: Test Spearman correlation, `design_hit` and `design_gain` as defined in Sec. III; with  $b = 10$  proposals `design_hit` takes values in multiples of 0.1 per run. We also log the fraction of all single mutants that beat the parent (`frac_mutants_better`), which is the exact expected hit rate of random proposals and is reported as “Chance” in Table I, and fit time. `design_gain` is logged in the registry but not analysed in this paper.

b) Further groups.: `sweep_n`:  $N \in \{250, 500, 2000, 4000\}$  on `L15_A4_K2` and `L20_A20_K1` for the four learned systems (5 seeds, where the proposal had planned 3;  $N = 1000$  from the main group). `abl_pair`: the two pairwise-ridge ablations on the eight main tasks (5 seeds).

c) Tuning and compute.: All hyperparameters are selected inside each run on training data only (Sec. III); the test set is used only for reporting; nothing was tuned across tasks. Hardware: a shared CPU laptop, 2 threads per run, two runs in parallel; according to the run registry the longest of the 440 runs took about 22 s and the registry spans about 14 minutes of wall-clock. Code, protocol and

the run registry are in the repository (method/run.py, experiments/PROTOCOL.md). With five seeds the smallest attainable one-sided Wilcoxon  $p$  is  $1/32 \approx 0.031$ ; we report effect sizes and seed spreads, and treat all  $p$ -values (paired  $t$ -tests from rh compare; Wilcoxon tests and rank correlations from analysis/hyp.py) as indicative.

## V. Results

Table I gives mean test Spearman and mean design\_hit over five seeds for the eight main tasks at  $N = 1000$ ; per-seed standard deviations of both endpoints are in Table II, and Fig. 1 plots both endpoints against  $K$ . Comparisons are paired over seeds. Paired  $t$ -tests come from rh compare (files results/tables/compare\_main\_\*.csv); the Wilcoxon tests, rank correlations and tests against the exact chance rate come from analysis/hyp.py (results/hypotheses.txt). With  $n = 5$  and no correction for multiple comparisons, all  $p$ -values are indicative only. All statements below are about simulated NK landscapes at two  $(L, A)$  settings in which  $L$  and  $A$  change together.

a) H1 (additive models absorb an additive landscape): supported.: At  $K = 0$  additive ridge reaches Spearman 1.000 at (15, 4) and 1.000 at (20, 20) (threshold 0.95). The MLP reaches 0.994 at L20\_A20\_K0, while the CNN is lower at 0.965 (Table I); at  $K = 0$  neither network improves on additive ridge. This is a sanity check on the simulator and the pipeline, not a finding.

b) H2 (epistasis hurts the additive model; pairs recover  $K=1$ ): supported, but weakly at (20, 20):. Additive Spearman falls from 1.000 to 0.596, 0.310, 0.079 for  $K = 0, 1, 2, 4$  at (15, 4), and from 1.000 to 0.110, 0.021, 0.014 at (20, 20). Pairwise ridge beats additive ridge at  $K = 1$  in all five seeds on both tasks (so the one-sided Wilcoxon test of analysis/hyp.py takes its smallest attainable  $p$ -value with five seeds,  $1/32$ ). At  $K = 1$  every contribution  $T_i$  is a function of two sites, so the model class contains the truth: at (15, 4) the pairwise model reaches 0.998 versus 0.596. At (20, 20) the gain is only 0.138 versus 0.110. A plausible reason is that 76,000 pair coefficients must be estimated from 1000 sequences; this is consistent with, but not isolated by, the  $N$  sweep and the oracle-graph ablation of Sec. VI.

c) H3 (small networks do not beat pairwise ridge at this scale): supported at  $N = 1000$ :. Neither the CNN nor the MLP exceeds pairwise ridge in mean Spearman in any of the six cells with  $K \geq 1$  (Table I). The networks have higher means than additive ridge on L15\_A4\_K1 and L15\_A4\_K2 (0.746/0.815 and 0.384/0.383 versus 0.596 and 0.310 for CNN/MLP/Additive; at  $K = 2$  the differences are within the seed SDs of Table II), i.e. at  $K = 1$  they absorb some non-additivity, but less than the model that has pair features built in. At (20, 20) their means are at or below additive ridge. The CNN’s locality prior may be a poor match for the random-neighbour structure of these landscapes; we did not test this, nor

other architectures, so this is not a statement about CNNs in general.

d) H4 (design success tracks prediction): registered criterion met, evidence weak.: The registered test is that, across the 40 (task, seed) units, the Spearman correlation between the Pairwise–Additive gain in test Spearman and the gain in design\_hit is positive. It is positive (results/hypotheses.txt, Fig. 2). Its nominal  $p$ -value treats the 40 units as independent, which they are not (five seeds in each of eight tasks), and the association is carried by one cell, L15\_A4\_K1, where pairwise ridge reaches design\_hit 0.90 versus 0.38 for additive ridge (Table III): without that cell the correlation is smaller and not nominally significant, and over the eight task means it is not nominally significant either (results/hypotheses.txt). On L15\_A4\_K2, a Spearman gain from 0.310 to 0.508 comes with a design\_hit change from 0.10 to 0.18 only (paired  $t$ -test  $p = 0.34$ , Table III), within the seed spread of Table II. So prediction gains translated into a clear design gain in one cell only.

The second part of the registered test holds for the means: at  $K \leq 1$  every learned model’s mean design\_hit exceeds Random’s and the exact chance rate (Table I). The margin is large at  $K = 0$  and for pairwise ridge at L15\_A4\_K1. Elsewhere it is not: at L20\_A20\_K1 all four learned models (0.30 to 0.38 versus 0.24 for Random and 0.233 for chance) are within their seed SDs (0.15 to 0.28), and at L15\_A4\_K1 the additive, CNN and MLP margins over the Random system are not distinguishable from noise in paired  $t$ -tests (rh compare), although the CNN and MLP are nominally above the exact chance rate there (results/hypotheses.txt).

At  $K \geq 2$  no learned model’s design\_hit differs from Random’s in paired  $t$ -tests (rh compare with Random as the reference, results/tables/compare\_main\_design\_hit\_ref\_random.csv; four tasks, four models; the unpaired Welch test in the same output is nominally at the conventional threshold for additive and pairwise ridge at L20\_A20\_K2,  $p = 0.05$  for additive ridge). One pattern in the means is left unexplained: at L20\_A20\_K2 and L20\_A20\_K4 additive ridge has design\_hit 0.34 and 0.18, against 0.14 and 0.08 for Random and 0.185 and 0.115 for chance, while its Spearman on random test sequences is low (0.021, 0.014; Random 0.007, 0.008). With five seeds of 10 proposals these margins are not distinguishable from noise (paired  $t$ -test  $p = 0.15$  and 0.30 against Random, Table III; the tests against the exact chance rate in results/hypotheses.txt are not nominally significant either), so we draw no conclusion from them. Whether a model can rank the single mutants of one parent while its Spearman on random, far-apart sequences is near zero would need more seeds and larger proposal batches to test.

e) Where every model fails.: At  $K = 4$  for both settings and at (20, 20) with  $K \geq 2$ , all four systems have

TABLE I

Simulated NK landscapes,  $N = 1000$  training variants, 5000 test variants, mean over 5 seeds. Left: held-out Spearman; right: design\_hit (fraction of the 10 proposed single mutants of the best measured sequence that beat it). Chance: mean fraction of all single mutants of the parent that beat it, i.e. the exact expected hit rate of random proposals, of which the Random column is a noisy 10-proposal estimate. Bold: best of the four learned systems (Random excluded; ties in bold).

| Task       | Spearman |       |       |       |        | design_hit |       |      |      |       | Chance |
|------------|----------|-------|-------|-------|--------|------------|-------|------|------|-------|--------|
|            | Add.     | Pair. | CNN   | MLP   | Rand.  | Add.       | Pair. | CNN  | MLP  | Rand. |        |
| L15_A4_K0  | 1.000    | 1.000 | 0.999 | 0.999 | 0.003  | 0.98       | 0.98  | 0.92 | 0.94 | 0.32  | 0.258  |
| L15_A4_K1  | 0.596    | 0.998 | 0.746 | 0.815 | 0.002  | 0.38       | 0.90  | 0.40 | 0.52 | 0.28  | 0.213  |
| L15_A4_K2  | 0.310    | 0.508 | 0.384 | 0.383 | -0.000 | 0.10       | 0.18  | 0.18 | 0.22 | 0.14  | 0.102  |
| L15_A4_K4  | 0.079    | 0.080 | 0.065 | 0.063 | -0.001 | 0.10       | 0.08  | 0.08 | 0.08 | 0.06  | 0.067  |
| L20_A20_K0 | 1.000    | 0.995 | 0.965 | 0.994 | -0.007 | 1.00       | 1.00  | 1.00 | 1.00 | 0.38  | 0.263  |
| L20_A20_K1 | 0.110    | 0.138 | 0.077 | 0.090 | 0.009  | 0.34       | 0.38  | 0.30 | 0.32 | 0.24  | 0.233  |
| L20_A20_K2 | 0.021    | 0.025 | 0.013 | 0.020 | 0.007  | 0.34       | 0.30  | 0.14 | 0.16 | 0.14  | 0.185  |
| L20_A20_K4 | 0.014    | 0.014 | 0.006 | 0.013 | 0.008  | 0.18       | 0.14  | 0.04 | 0.16 | 0.08  | 0.115  |

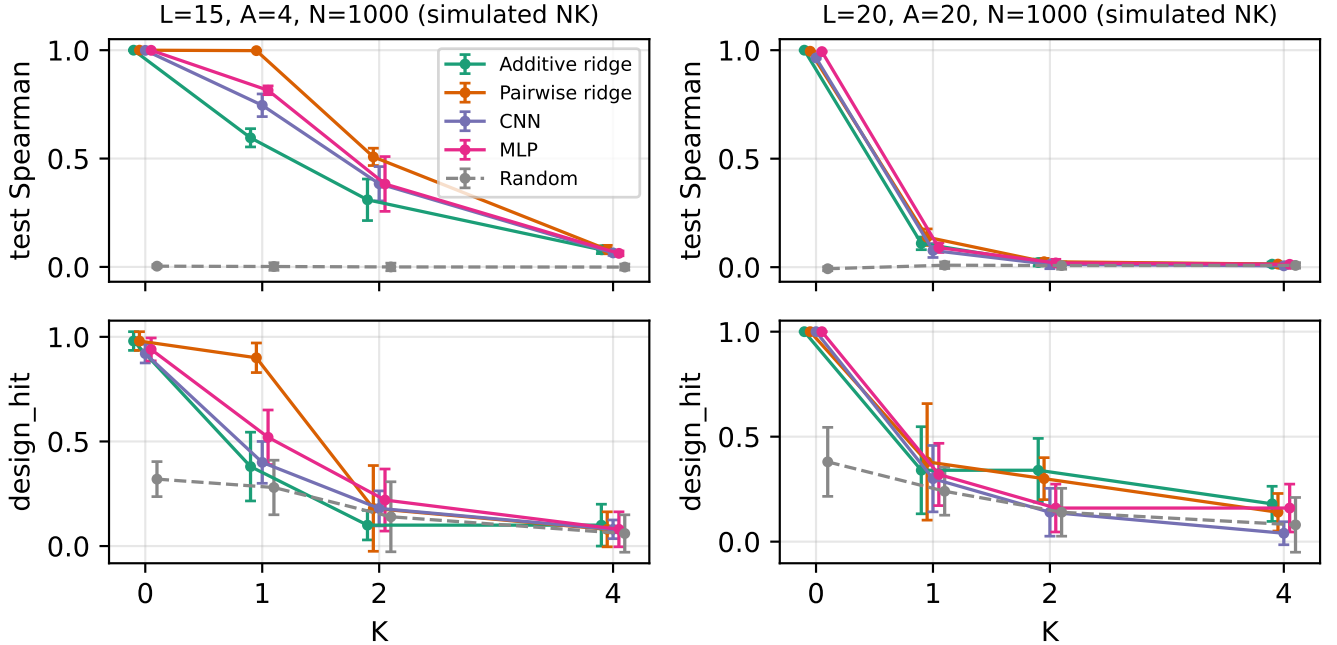


Fig. 1. Spearman (top) and design\_hit (bottom) versus  $K$  for the two  $(L, A)$  settings,  $N = 1000$ ; mean  $\pm$  SD over 5 seeds. Everything is simulated.

mean Spearman of at most 0.080 and, for  $(20, 20)$ , at most 0.025 (Table I); differences between system means there are comparable to or smaller than the seed SDs (Table II), and on L20\_A20\_K4 all learned means ( $\leq 0.014$ ) are close to Random's (0.008). The data do not say whether more data would help in these regimes; the  $N$  sweep only covers L15\_A4\_K2 and L20\_A20\_K1. Also, pairwise ridge is not the best system at  $K = 0$  at  $(20, 20)$  (0.995 versus 1.000), which may be a small cost of fitting uninformative pair features: the oracle-graph variant, which has no pair features at  $K = 0$ , reaches 1.000 (Table V).

## VI. Ablations and Analysis

a) Training-set size (H5): supported.: Table IV and Fig. 3 vary  $N$  from 250 to 4000 on L15\_A4\_K2

and L20\_A20\_K1 ( $N = 1000$  comes from the main group). The registered test concerns the  $K = 2$  task: the Pairwise–Additive Spearman gap there is larger at  $N = 4000$  (0.669 versus 0.358) than at  $N = 250$  (0.301 versus 0.234), and grows at every step in between; on L20\_A20\_K1 it also grows (from 0.070 versus 0.065 to 0.302 versus 0.211). Regarding H3, the networks stay below pairwise ridge at every  $N$  on both tasks. On L15\_A4\_K2 they are indistinguishable from additive ridge at  $N = 500$  (0.284, 0.283 versus 0.285) and close the distance to pairwise ridge at  $N = 4000$  (0.616 and 0.628 versus 0.669). On L20\_A20\_K1, however, they are at or below additive ridge at every  $N$  (the one exception, MLP 0.070 versus 0.068 at  $N = 500$ , is far inside the seed spread of Fig. 3), the best score at  $N = 4000$  is only 0.302,

TABLE II  
Standard deviation over the 5 seeds of test Spearman and of design\_hit (same cells as Table I).

| Task       | SD of Spearman |       |       |       | SD of design_hit |       |      |      |       |
|------------|----------------|-------|-------|-------|------------------|-------|------|------|-------|
|            | Add.           | Pair. | CNN   | MLP   | Add.             | Pair. | CNN  | MLP  | Rand. |
| L15_A4_K0  | 0.000          | 0.000 | 0.000 | 0.000 | 0.04             | 0.04  | 0.04 | 0.05 | 0.08  |
| L15_A4_K1  | 0.042          | 0.002 | 0.052 | 0.020 | 0.16             | 0.07  | 0.10 | 0.13 | 0.13  |
| L15_A4_K2  | 0.096          | 0.040 | 0.079 | 0.126 | 0.07             | 0.20  | 0.08 | 0.15 | 0.17  |
| L15_A4_K4  | 0.017          | 0.019 | 0.011 | 0.011 | 0.10             | 0.08  | 0.04 | 0.08 | 0.09  |
| L20_A20_K0 | 0.000          | 0.000 | 0.005 | 0.002 | 0.00             | 0.00  | 0.00 | 0.00 | 0.16  |
| L20_A20_K1 | 0.029          | 0.038 | 0.032 | 0.022 | 0.21             | 0.28  | 0.16 | 0.15 | 0.11  |
| L20_A20_K2 | 0.016          | 0.015 | 0.020 | 0.017 | 0.15             | 0.10  | 0.11 | 0.11 | 0.11  |
| L20_A20_K4 | 0.010          | 0.008 | 0.011 | 0.011 | 0.08             | 0.09  | 0.05 | 0.11 | 0.13  |

TABLE III  
H4 details: mean paired difference in design\_hit over 5 seeds (reference minus the other system) with a two-sided paired  $t$ -test, from rh compare with additive ridge as the reference.  $p$ -values are nominal and uncorrected. The rank correlations of the registered test and the tests against the exact chance rate are in results/hypotheses.txt.

| Quantity                   | Value | $p$   |
|----------------------------|-------|-------|
| Add.-Pair. hit, L15_A4_K1  | -0.52 | 0.003 |
| Add.-Pair. hit, L15_A4_K2  | -0.08 | 0.34  |
| Add.-Pair. hit, L20_A20_K1 | -0.04 | 0.48  |
| Add.-Rand. hit, L20_A20_K2 | 0.20  | 0.15  |
| Add.-Rand. hit, L20_A20_K4 | 0.10  | 0.30  |

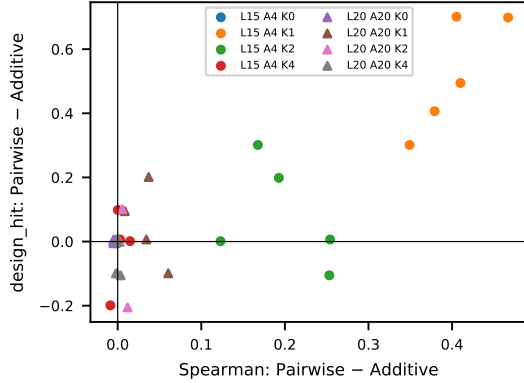


Fig. 2. Per (task, seed): gain of Pairwise over Additive in Spearman (x) versus in design\_hit (y; small vertical jitter added for display). Circles:  $L = 15, A = 4$ ; triangles:  $L = 20, A = 20$ .

and their distance to pairwise ridge does not shrink (0.162 and 0.192 versus 0.302). So H3 held at every  $N$  we ran; the sweep suggests the gap may close with more data on L15\_A4\_K2, but  $N > 4000$  was not run and only two tasks were swept.

b) Pair-feature ablations.: Table V compares tuned Pairwise ridge with (i) the penalty ratio fixed at  $r = 1$  and (ii) pair features restricted to the true interaction graph (an oracle). Fixing  $r = 1$  matters only where pair features carry no signal: at L20\_A20\_K0 it costs  $0.995 \rightarrow 0.897$ ; elsewhere the differences are within one

TABLE IV  
Test Spearman versus training size  $N$  (mean of 5 seeds).

| Task       | $N$  | Add.  | Pair. | CNN   | MLP   |
|------------|------|-------|-------|-------|-------|
| L15_A4_K2  | 250  | 0.234 | 0.301 | 0.232 | 0.234 |
| L15_A4_K2  | 500  | 0.285 | 0.374 | 0.284 | 0.283 |
| L15_A4_K2  | 1000 | 0.310 | 0.508 | 0.384 | 0.383 |
| L15_A4_K2  | 2000 | 0.335 | 0.594 | 0.515 | 0.517 |
| L15_A4_K2  | 4000 | 0.358 | 0.669 | 0.616 | 0.628 |
| L20_A20_K1 | 250  | 0.065 | 0.070 | 0.048 | 0.058 |
| L20_A20_K1 | 500  | 0.068 | 0.091 | 0.060 | 0.070 |
| L20_A20_K1 | 1000 | 0.110 | 0.138 | 0.077 | 0.090 |
| L20_A20_K1 | 2000 | 0.163 | 0.215 | 0.115 | 0.140 |
| L20_A20_K1 | 4000 | 0.211 | 0.302 | 0.162 | 0.192 |

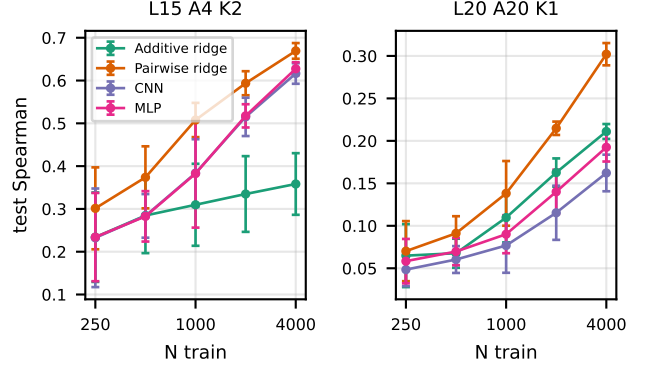


Fig. 3. Test Spearman versus  $N$  (mean  $\pm$  SD, 5 seeds).

seed standard deviation. The oracle graph helps in two cells:  $0.508 \rightarrow 0.621$  on L15\_A4\_K2 and  $0.138 \rightarrow 0.250$  on L20\_A20\_K1; it is at ceiling (1.000) at  $K \leq 1$  for (15,4). Even then L15\_A4\_K2 stays at 0.621, below the level that  $K = 1$  reaches, which is consistent with (but not proof of) order-3 terms that no pairwise model can capture; and at  $K = 4$  nothing helps (0.084 at best). We did not run a three-way-interaction model, so the order-3 explanation is untested. Knowing which pairs interact thus raised mean Spearman clearly in those two cells and changed it little in the other six (Table V); this is an

TABLE V  
Pairwise-ridge ablations, test Spearman, mean  $\pm$  SD over 5 seeds  
( $N = 1000$ ).

| Task       | Pair. (tuned $r$ ) | Pair. ( $r=1$ )   | Pair. (oracle graph) |
|------------|--------------------|-------------------|----------------------|
| L15_A4_K0  | 1.000 $\pm$ 0.000  | 1.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000    |
| L15_A4_K1  | 0.998 $\pm$ 0.002  | 0.998 $\pm$ 0.002 | 1.000 $\pm$ 0.000    |
| L15_A4_K2  | 0.508 $\pm$ 0.040  | 0.500 $\pm$ 0.048 | 0.621 $\pm$ 0.024    |
| L15_A4_K4  | 0.080 $\pm$ 0.019  | 0.079 $\pm$ 0.031 | 0.084 $\pm$ 0.028    |
| L20_A20_K0 | 0.995 $\pm$ 0.000  | 0.897 $\pm$ 0.007 | 1.000 $\pm$ 0.000    |
| L20_A20_K1 | 0.138 $\pm$ 0.038  | 0.150 $\pm$ 0.025 | 0.250 $\pm$ 0.024    |
| L20_A20_K2 | 0.025 $\pm$ 0.015  | 0.028 $\pm$ 0.017 | 0.029 $\pm$ 0.013    |
| L20_A20_K4 | 0.014 $\pm$ 0.008  | 0.013 $\pm$ 0.007 | 0.015 $\pm$ 0.008    |

upper reference for what pair selection could add in this simulator, obtained with information a practitioner lacks.

## VII. Limitations

- Simulated, i.i.d. NK only. Contributions are i.i.d. uniform tables with random neighbours; real proteins have correlated effects, global epistasis, sparsity, noise and structured neighbourhoods. The results say how these models behave on this family and nothing about proteins.
- Scale. Eight tasks at  $N = 1000$ , five seeds, two  $(L, A)$  pairs,  $K \in \{0, 1, 2, 4\}$ ;  $K = 3$ , intermediate alphabets and noisy labels were not run.  $L$  and  $A$  change together between the two settings, so no difference between them can be attributed to alphabet size or to length alone. The  $N$  sweep covers two tasks and none of the regimes where every model fails. Per-cell differences with five seeds are often within seed variation.
- Reimplemented, lightly and unequally tuned baselines. The CNN and MLP have one architecture each and 2 configurations (weight decay) against 24 for pairwise ridge and 6 for additive ridge, and they are fit on 85% of the training set without the refit on all data that the ridge models get. Both asymmetries favour ridge; larger networks, a refit, ensembles, other regularisers or pretraining could change the net-versus-ridge comparison (H3).
- Test distribution. Random train and test sequences; no homology, no extrapolation split, no noise.
- Design endpoint. One greedy step of 10 proposals from one parent; design\_hit is quantised in tenths per run, so with five seeds only large differences are detectable: outside  $K = 0$  and pairwise ridge at L15\_A4\_K1, the design\_hit differences in this paper are at most nominally significant in uncorrected tests, and the H4 correlation rests on that one cell. The parent is the best training sequence (selected on noiseless fitness), and no closed-loop or batch-diversity design was tried.

## VIII. Conclusion

On simulated random-neighbour NK landscapes at two  $(L, A)$  settings, additive ridge absorbs  $K = 0$  completely,

and pairwise ridge absorbs  $K = 1$  at  $(15, 4)$  (0.998 Spearman at  $N = 1000$ ) but not at  $(20, 20)$  (0.138); since only these two settings were run, the difference cannot be attributed to alphabet size rather than length. At  $K = 4$ , and at  $(20, 20)$  with  $K \geq 2$ , no model learned much from  $N = 1000$  (mean Spearman at most 0.080); larger  $N$  was not tested in those regimes. The small CNN and MLP did not beat pairwise ridge at  $N = 1000$  and stayed below it up to  $N = 4000$  on the two sweep tasks, with a narrowing gap on L15\_A4\_K2 only. The registered design criterion was met in sign, but the association between prediction gains and single-step design gains rests on one task cell, and with 10 proposals and five seeds the design hit rates at  $K \geq 2$  could not be reliably separated from chance. A fuller study would add more seeds and larger proposal batches for the design endpoint, intermediate  $K$ ,  $L$  and  $A$  varied separately, noisy labels, correlated (non-i.i.d.) landscapes, stronger neural baselines, larger  $N$  in the hard regimes, and real assays; none of that was done here.

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